

Cross-Modal 3D Brain MRI Re-Identification

A training-free, content-honest retrieval pipeline

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The problem

Match a patient across two MRI modalities. Given a contrast T1 (**ceT1**) query, find the *same patient's* T2 scan in a gallery.

- Same brain, **different physics** — intensities don't correspond.
- No labels to train on (~ 180 pairs). **Training-free** required.
- Score: **MRR**, rank-1 dominated, macro-averaged over 3 datasets.



Three regimes, increasing difficulty

dataset1

co-registered
(shared voxel grid)
content already lines up

dataset2

independent
rigid + elastic
correspondence **destroyed** $\Rightarrow$
must re-align

dataset3

preop $\rightarrow$ intraop
+ tumour resection
tissue **removed** + a planted
geometric leak

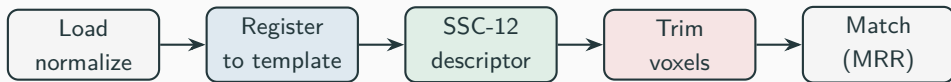
One pipeline; each regime turns on a different lever.

Methodology — four deterministic steps

1. **Register to a neutral template** — Mutual Information, *modality-blind* (rigid → affine cascade). fixes geometry
2. **Describe with SSC-12** — 12-edge self-similarity context. *Modality-invariant*: encodes local structure, not intensity. fixes modality
3. **Trim mismatched voxels** — drop the worst $X\%$ of residuals. fixes resection
4. **Assign** — greedy (deployable) or Hungarian (batch).

No network, no training, no labels — **deterministic and auditable**.

Architecture



Per-regime configuration of the *same* pipeline:

	register	descriptor	trim
dataset1	— (shared grid)	SSC-12	light
dataset2	affine cascade	SSC-12	light (0.5)
dataset3	affine cascade	SSC-12	heavy (0.75)

Key finding — we found a leak, and refused it

dataset3 had a **planted geometric co-location**: the true target was resampled into the query's exact voxel space $\Rightarrow$ a perfect overlap with **zero anatomy**.

use the leak (resample)	1.000	geometry, not the patient
break it, don't re-align	0.259	proves the 1.0 was the grid
break it + re-register on content	0.741	re-earned from anatomy

We measured the leak's exact worth (+0.74), then declined it.

Results (real leaderboard)

Setting	MRR
Leak (<i>declined</i>)	1.000
Honest, full (with assignment)	0.930
Honest, deployable (single-query)	0.725
dataset1 / dataset2 / dataset3 (honest)	1.0 / strong / 0.74

Honest caveat (we say it first): the $0.930 - 0.725 = +0.205$ gap is the Hungarian *assignment* exploiting the eval bijection — a disclosed *batch* optimization, not deployable single-query anatomy. The deployable number is **0.725**.

What we ruled out

~22 techniques tested, every dead end documented (hypothesis → result → reason):

- 8 classifiers (incl. TabPFN), learned 3D encoder
- feature fusion, multi-scale, low-rank
- NGF / NMI / shape / radiomics descriptors
- Bayesian search, GA, DP / DTW alignment
- matrix methods, Markov / diffusion re-rank
- deformable registration (over-normalizes)

The bottleneck is the content signal — not the model.

One positive add-on: Bayesian calibration ⇒ a deploy-time abstention layer.

Takeaways

Method — SSC + content-registration + trim.

Training-free, deterministic, auditable.

Integrity — found the leak, measured it ($1.0 \rightarrow 0.26$),
re-earned 0.74 from anatomy.

Rigor — mapped the whole frontier; ruled out 22
alternatives with reasons.